PYTHON'S MAGIC TOOLBOXES

Modules & The Standard Library

GRADE 4 | SESSION 1

PLANRS Academy

Lesson 1: Imagine a Masala Dabba!

✨ The Secret of the Spice Box

Think about a Masala Dabba in your kitchen. It is a special toolbox for cooking!

📸 ***Visual:*** *A colorful Indian Masala Dabba with various spices.*

- It has all the spices you need to cook tasty food!
- You don't grow spices every time you want to eat; you just take what you need from the box!

In Python, we have something exactly like this called a **Module**.

Understanding Modules

⚙️ What is a Python Module?

A module is a special toolbox in Python that contains ready-made tools (code) to help us build amazing things.

Just like the Masala Dabba, we only 'import' the tools we need for our specific recipe (or program).

Meet Our Friends:

- **Aarav** loves numbers and uses the math module to solve big problems!
- **Diya** uses modules to measure her garden!

How to use 'import'

To use a module, we use the magic word: `import`.

```
import math
print(math.sqrt(16)) # Finding the square root!
```

This line brings the entire math toolbox into your program!

The Math & Random Modules

📐 The Math Module

The math module helps us with special calculations that would be hard to do by ourselves.

For example, Arjun uses `math.pi` to calculate the perfect area of a Rangoli circle!

> 🎨 ***Visual:*** *A child creating a geometric Rangoli pattern on the floor.*

🎲 The Random Module

The random module is all about making choices and having fun!

Gully Cricket Toss: Ever argued about who bats first? Use `random.choice()` for a digital coin toss!

Time to Code!

🏏 Activity 1: Gully Cricket Toss

Let's code a program to pick the first batter!

```
import random
players = ['Aarav', 'Diya', 'Arjun']
print(random.choice(players), "bats first!")
```

Try it: Add your best friends' names to the list!

🏡 Activity 2: Rangoli Garden

Diya needs to find the side length of a square garden with an area of 81 square meters.

```
import math
area = 81
side = math.sqrt(area)
print("The side length is", side, "meters")
```

The Standard Library & Summary

🚌 The Standard Library

Python comes with many, many toolboxes already built-in. This collection is called the **Standard Library**.

> **🚌 The School Bus Analogy:**
>
> It is like a First Aid Kit in every Indian school bus. You don't need to build it; it is just there ready to help you anytime!

❓ Quick Quiz!

1. What is a Python module?
2. How do we bring a module into our program?
3. Name one thing the math or random module can do!

Great job today! See you in Session 2 for more magic!

PLANRS Academy

Student Practice Workbook (Grade 4)

Time to practice your Python superpowers! Answer these questions based on what you learned about magic toolboxes.

1. What everyday kitchen item is used in this lesson to explain Python modules?

---

2. What is a "module" in Python? Explain it in your own words.

---

3. What is the magic keyword we write in our code to bring a toolbox into Python?

---

4. Which Python module should Aarav use if he wants to solve big math problems?

---

5. If you want to make a digital coin toss for a game, which module would you import?

---

6. Look at this code: `math.sqrt(16)`. What specific math problem is this tool solving?

---

7. What does the tool `random.choice()` do when we give it a list of names?

---

8. What is the "Standard Library" in Python?

---

9. What analogy does the book use to explain the Standard Library? (Hint: Think about a school vehicle!)

---

10. True or False: You need to build a first aid kit or write the math code from scratch every time you use Python.

---

11. Why did Arjun use `math.pi` in his program? What was he creating?

---

12. If Diya's square garden has an area of 36 square meters, what command would find the side length?

---

13. In Activity 1, what happens if you add your own name to the `players` list?

---

14. Why is it helpful to import only the tools we need instead of making everything from scratch?

---

15. Write a line of code that imports the random module.
